

A finite Toeplitz counterexample to the operator-norm upgrade in arXiv:2506.01610

Agent lane 19

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Source Question

The source paper is:

S. Finski, *Bernstein-Markov measures and Toeplitz theory*, arXiv:2506.01610.

The relevant passage is Theorem 1.5 and Remark 1.6. Theorem 1.5 proves that, in the Bernstein–Markov setting of the paper, Toeplitz operators form an algebra under composition in normalized Schatten norms:

$$\lim_{k \rightarrow \infty} \|T_k(f) \circ T_k(g) - T_k(fg)\|_p = 0, \quad 1 \leq p < \infty.$$

Remark 1.6 then recalls that in the classical smooth positive volume-form setting the analogous statement holds with the operator norm in place of the normalized Schatten norm, and says that it remains open whether analogous results hold in the paper’s more general setting.

Theorem 1.5. *The space of Toeplitz operators forms an algebra under composition, and the symbol map is an algebra morphism when $\mathcal{C}^0(K')$ is endowed with the standard multiplication. In particular for any $f, g \in \mathcal{C}^0(K)$, $p \in [1, +\infty[$, we have*

$$\lim_{k \rightarrow \infty} \|T_k(f) \circ T_k(g) - T_k(f \cdot g)\|_p = 0. \quad (1.10)$$

Remark 1.6. Bordemann–Meinrenken–Schlichenmaier in [14] showed that when μ is a volume form on X , L is an ample line bundle and h^L is a smooth metric with a strictly positive curvature, the analogous statement holds if the p -Schatten norm in Definition 1.3 is replaced by the operator norm. The author in [31] proved that under the same assumptions, Theorem 1.5 remains valid when the space $\mathcal{C}^0(X)$ in Definition 1.3 is replaced by $L^\infty(X)$. It remains open if in our setting it is possible to get the analogues of the above results.

It would be interesting to identify natural conditions on μ that allow for a refinement of (1.10) to include lower-order terms in k , and ultimately to analyze the commutator $\frac{1}{k}[T_k(f), T_k(g)]$ and its connections to symplectic geometry, particularly in light of the foundational works [14] and

Result

Theorem 1. *The operator-norm analogue of Theorem 1.5 is false in the general Bernstein–Markov setting of arXiv:2506.01610.*

More precisely, in the circle model from Section 4 of the source paper, where Definition 1.3 recovers finite Toeplitz matrices, there is a real continuous symbol h such that

$$\|T_N[h]T_N[h] - T_N[h^2]\| = 1$$

for every $N \geq 2$. Hence this error does not converge to zero in operator norm, although it converges to zero in every normalized Schatten p -norm, $1 \leq p < \infty$.

Proof Intuition

The source theorem is normalized by the dimension. A fixed-rank boundary error therefore disappears in every normalized Schatten norm. The operator norm does not see dimension averaging, so the same boundary error remains of size one.

Classical finite Toeplitz matrices give exactly such a boundary error. The symbol $2 \cos \theta$ quantizes to a truncated shift plus its adjoint. Squaring that finite matrix almost gives the Toeplitz matrix of $4 \cos^2 \theta$, but the two endpoints of the finite basis have no predecessor or successor. The missing neighbors are precisely two rank-one endpoint projections.

Proof

We use the finite Toeplitz-matrix realization explicitly described in Section 4 of the source paper. Let $N \geq 2$, and identify the relevant section space with $\mathbb{C}^N$ with orthonormal basis $e_0, \dots, e_{N-1}$. If

$$a(e^{i\theta}) = \sum_{m \in \mathbb{Z}} a_m e^{im\theta}$$

is a trigonometric polynomial, the associated Toeplitz matrix is

$$T_N[a] = (a_{i-j})_{0 \leq i, j \leq N-1}.$$

This is the matrix displayed in Section 4 of the source paper.

Let S_N be the truncated unilateral shift

$$S_N e_j = \begin{cases} e_{j+1}, & 0 \leq j \leq N-2, \\ 0, & j = N-1. \end{cases}$$

For the real continuous symbol

$$h(e^{i\theta}) = e^{i\theta} + e^{-i\theta} = 2 \cos \theta,$$

the only nonzero Fourier coefficients are $h_1 = h_{-1} = 1$. Hence

$$T_N[h] = S_N + S_N^*.$$

Also,

$$h^2(e^{i\theta}) = e^{2i\theta} + 2 + e^{-2i\theta},$$

so

$$T_N[h^2] = S_N^2 + (S_N^*)^2 + 2I_N.$$

On the other hand,

$$T_N[h]^2 = (S_N + S_N^*)^2 = S_N^2 + S_N S_N^* + S_N^* S_N + (S_N^*)^2.$$

Therefore

$$T_N[h]^2 - T_N[h^2] = S_N S_N^* + S_N^* S_N - 2I_N.$$

Writing P_0 and P_{N-1} for the rank-one projections onto $\mathbb{C}e_0$ and $\mathbb{C}e_{N-1}$, respectively, we have

$$S_N S_N^* = I_N - P_0, \quad S_N^* S_N = I_N - P_{N-1}.$$

Thus

$$T_N[h]^2 - T_N[h^2] = -(P_0 + P_{N-1}).$$

Since P_0 and P_{N-1} are nonzero orthogonal projections,

$$\|T_N[h]^2 - T_N[h^2]\| = 1$$

for every $N \geq 2$.

By contrast, for the normalized Schatten p -norm used in Definition 1.3 of the source paper,

$$\|A\|_p = \left(\frac{1}{N} \operatorname{Tr} |A|^p \right)^{1/p},$$

the same defect has only two nonzero singular values, both equal to 1. Therefore

$$\|T_N[h]^2 - T_N[h^2]\|_p = \left(\frac{2}{N} \right)^{1/p} \rightarrow 0.$$

This is exactly the phenomenon allowed by Theorem 1.5 and exactly the phenomenon excluded by the proposed operator-norm analogue. The general operator-norm upgrade is therefore false.

Verification Notes

The example is inside the source paper's general Bernstein–Markov framework: Section 4 states that for $X = \mathbb{P}^1$, $L = \mathcal{O}(1)$, and Lebesgue measure on a great circle, Definition 1.3 recovers the classical Toeplitz matrices. The chosen symbol $h = 2 \cos \theta$ is real and continuous, so it is admissible even under a self-adjoint real-symbol reading.

The proof uses only the displayed finite Toeplitz matrix convention and the elementary algebra of the truncated shift. No asymptotic estimate or external operator theory theorem is needed.

Novelty Check and Limitations

Local index searches for 2506.01610, Bernstein–Markov, Toeplitz theory, operator norm, and Toeplitz matrices found no prior packet for this source question. A bounded web search for the exact source arXiv id, the paper title, and phrases around the operator-norm Toeplitz product did not find a later paper explicitly answering Remark 1.6.

This packet disproves the operator-norm analogue in the full Bernstein–Markov setting. It does not rule out operator-norm product formulas under additional regularity assumptions on μ , and it does not settle the separate L^∞ -symbol analogue mentioned in the same remark.

Human Review Recommendation

Check that the intended reading of Remark 1.6 includes the full Bernstein–Markov setting of Theorem 1.5 rather than only an unspecified future subclass of measures. Under that natural reading, the counterexample is complete and the computation is exact.

References

- [1] S. Finski, *Bernstein-Markov measures and Toeplitz theory*, arXiv:2506.01610, 2025.
- [2] U. Grenander and G. Szego, *Toeplitz Forms and Their Applications*, 2nd ed., Chelsea, 1984.